

[Siddhant Aggarwal]

AI Engineer — Machine Learning Researcher

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LinkedIn — GitHub — Portfolio

Education

Master of Engineering in Information Technology

SRH Hochschule Heidelberg, Germany

Specialization: Artificial Intelligence

Expected Graduation: March 2027

GPA: 1.9 — **Focus:** Deep Learning, Computer Vision, Generative AI, Uncertainty Quantification

Bachelor of Engineering in Computer Engineering

Thapar Institute, India

Graduated: June 2024

Technical Skills

Programming: Python (Expert), SQL, C++ (Academic), Bash, Git

Deep Learning: PyTorch (Advanced), TensorFlow, CNNs, LSTMs, Transformers, Uncertainty Quantification

Generative AI: Langchain, LLMs (Llama), Fine-Tuning (LoRA/PEFT), Prompt Engineering, Hugging Face Transformers

Computer Vision: Image Classification, Medical Imaging, CNN Architecture Design, Feature Engineering

Classical ML: Scikit-learn, Random Forests, PCA, Hyperparameter Optimization (Optuna)

Data Science: Pandas, NumPy, Matplotlib, Seaborn, Statistical Analysis, Time-Series Forecasting

MLOps & Tools: Docker, Linux, FastAPI, Cloud Deployment (Hugging Face, AWS), Streamlit, Jupyter

Key Projects

CNN Architecture Research for Medical Imaging

Python, PyTorch, NumPy, Linux

- Conducted systematic research on CNN depth impact, training 20 experiments (4 architectures $\times$ 5 seeds) on 260K+ histopathology images (PatchCamelyon) for lymph node metastasis detection
- Achieved 90.30% ROC-AUC with 8-layer architecture (583K params), discovering sharp diminishing returns beyond 8 layers—24-layer model showed only +0.11% improvement despite $7\times$ more parameters
- Quantified training instability: variance increased $2.5\times$ with depth (1.13% $\rightarrow$ 2.87% std dev), empirically validating ResNet degradation problem through multi-seed statistical analysis
- Built custom GPU workstation and reproducible experiment tracking pipeline; recommended 8-layer model for clinical deployment (87% fewer parameters, optimal performance-efficiency trade-off)

LLM Fine-Tuning for Automated Content Generation

Python, PyTorch, LoRA, FastAPI

- Fine-tuned Llama-3.2 (3B parameters) using LoRA for structured text generation, applying prompt engineering to achieve consistent outputs
- Built end-to-end ML system: curated training data, implemented parameter-efficient fine-tuning, developed production API (FastAPI), deployed on cloud (Hugging Face Spaces)
- Applied 4-bit quantization for resource-constrained deployment, demonstrating practical generative AI application from research to production

Battery Health Prediction with Uncertainty Quantification

Python, PyTorch, Optuna

- Developed deep learning models for Remaining Useful Life (RUL) prediction using multi-modal sensor data (voltage, current, temperature) from NASA/CALCE datasets
- Implemented Monte Carlo Dropout for uncertainty quantification, providing confidence-aware predictions for reliable maintenance decision-making
- Conducted comparative analysis of Transformer vs. LSTM architectures, using automated hyperparameter optimization (Optuna) to identify optimal configurations
- Built robust preprocessing pipeline handling real-world sensor noise, drift, and missing data; evaluated using domain-appropriate metrics (RMSE, MAE, early prediction accuracy)

Professional Experience

Data Analytics Intern

Centre for Railway Information Systems (CRIS), India

Jan 2024 – June 2024

- Automated data processing pipelines using Python and SQL in Linux environments, reducing manual effort by 40%
- Cleaned and structured large-scale operational datasets, optimizing them for downstream machine learning workflows
- Developed version-controlled scripts (Git) ensuring reproducibility; collaborated with cross-functional teams to translate business requirements into technical solutions

Additional Information

Languages: English (Fluent), German (A2 - Improving), Hindi (Native)

Research Interests: Generative AI, Medical Imaging, Uncertainty Quantification, Trustworthy AI, Edge AI